

EXTENSION 2

Stage 6

Vectors (Ext2), V1 Working With Vectors (Ext2)

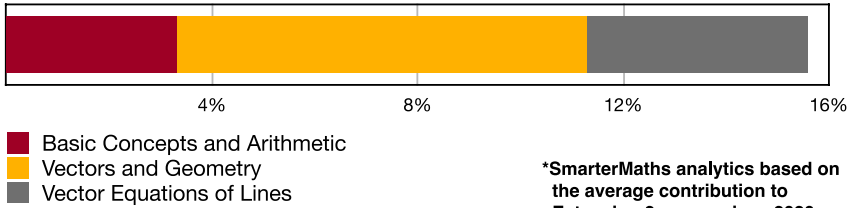
Vectors and Geometry

Teacher: Nardene Dodd

Exam Equivalent Time: 139.5 minutes (based on allocation of 1.5 minutes per mark)



V1 Working with Vectors



HISTORICAL CONTRIBUTION

- V1 Working With Vectors has contributed an impressive 15.5% per new syllabus Ext2 exam since it was introduced in 2020.
- We have split this topic into three categories for analysis purposes which are: 1-Basic Concepts and Arithmetic (3.3%), 2-Vectors and Geometry (8.0%) and 3-Vectors and Vector Equations of Lines (4.2%).
- This analysis looks at Vectors and Geometry.

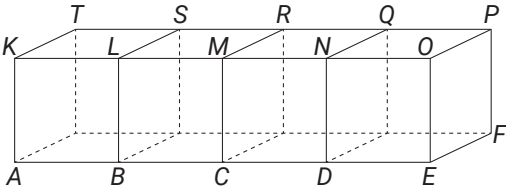
HSC ANALYSIS - What to expect and common pitfalls

- Vectors and Geometry (8.0%) is the new 600-pound Silverback in the Ext2 rainforest that beat its chest big time in 2023 with three longer answer questions for a total 9-mark allocation!
- The topic area has seen consistently low mean marks produced in new syllabus exams. Its importance in differentiating the best exam papers cannot be overstated.
- While the scope of questions examiners can ask is extremely broad, we have categorised them into 2D and 3D shapes which is a good starting point for structured revision in our view.
- The most common question type has looked at pyramids, both square (2021) and triangular (2023, 2022). A key revision focus area.
- 2D quadrilaterals and triangles were the subject of 2023 Ext2 11d and 2020 Ext2 15b and deserve attention.
- 2023 Ext2 15c and 2021 Ext2 16a looked at spheres and both caused major issues.

Questions

1. Vectors, EXT2 V1 2021 HSC 1 MC

Four cubes are placed in a line as shown on the diagram.



Which of the following vectors is equal to $\overrightarrow{AB} + \overrightarrow{CQ}$

- $\overrightarrow{AQ}$
- $\overrightarrow{CP}$
- $\overrightarrow{PB}$
- $\overrightarrow{RA}$

2. Vectors, EXT2 V1 2022 HSC 9 MC

Let $\mathbf{A}$ and $\mathbf{B}$ be two distinct points in three-dimensional space. Let $\mathbf{M}$ be the midpoint of $\mathbf{AB}$.

Let $\mathbf{S}_1$ be the set of all points $\mathbf{P}$ such that $\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$.

Let $\mathbf{S}_2$ be the set of all points $\mathbf{N}$ such that $|\overrightarrow{AN}| = |\overrightarrow{MN}|$.

The intersection of $\mathbf{S}_1$ and $\mathbf{S}_2$ is the circle $\mathbf{S}$.

What is the radius of the circle $\mathbf{S}$?

- $\frac{|\overrightarrow{AB}|}{2}$
- $\frac{|\overrightarrow{AB}|}{4}$
- $\frac{\sqrt{3}|\overrightarrow{AB}|}{2}$
- $\frac{\sqrt{3}|\overrightarrow{AB}|}{4}$

3. Vectors, EXT2 V1 2023 HSC 10 MC

Consider any three-dimensional vectors $\vec{a} = \vec{OA}$, $\vec{b} = \vec{OB}$ and $\vec{c} = \vec{OC}$ that satisfy these three conditions

$\vec{a} \cdot \vec{b} = 1$
 $\vec{b} \cdot \vec{c} = 2$
 $\vec{c} \cdot \vec{a} = 3.$

Which of the following statements about the vectors is true?

- A. Two of $\vec{a}$, $\vec{b}$ and $\vec{c}$ could be unit vectors.
- B. The points A , B and C could lie on a sphere centred at O .
- C. For any three-dimensional vector $\vec{a}$, vectors $\vec{b}$ and $\vec{c}$ can be found so that $\vec{a}$, $\vec{b}$ and $\vec{c}$ satisfy these three conditions.
- D. $\forall \vec{a}, \vec{b}$ and $\vec{c}$ satisfying the conditions, $\exists \vec{r}, \vec{s}$ and $\vec{t}$ such that $\vec{r}, \vec{s}$ and $\vec{t}$ are positive real numbers and $\vec{r}\vec{a} + \vec{s}\vec{b} + \vec{t}\vec{c} = \vec{0}$.

4. Vectors, EXT2 V1 EQ-Bank 14

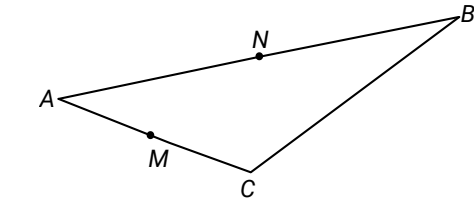
Use vector methods to find the coordinates of the point that divides the interval joining the points $A(7, -3, 0)$ and $B(2, 2, -10)$ in the ratio $2:3$. (3 marks)

5. Vectors, EXT2 V1 EQ-Bank 5

Use two vector methods to locate the midpoint of the interval joining the points $A(3, -2, 1)$ and $B(5, 4, -3)$. (3 marks)

6. Vectors, EXT2 V1 EQ-Bank 7

In triangle ABC , M is the midpoint of AC and N is the midpoint of AB .



Use vector methods to prove that

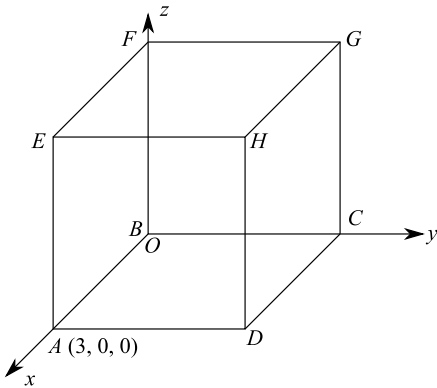
- i. $MN = \frac{1}{2}CB$ (2 marks)
- ii. MN is parallel to CB (1 mark)

7. Vectors, EXT2 V1 EQ-Bank 9

A parallelogram is formed by joining the points $P(-2, 1, 4)$, $Q(1, 4, 5)$, $R(0, 2, 3)$ and $S(a, b, c)$. Use vector methods to find a, b and c . (2 marks)

8. Vectors, EXT2 V1 SM-Bank 23

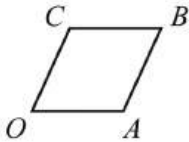
A cube with side length 3 units is pictured below.



- i. Calculate the magnitude of vector $\vec{AG}$. (1 mark)
- ii. Find the acute angle between the diagonals $\vec{AG}$ and $\vec{BH}$. (3 marks)

9. Vectors, EXT2 V1 2015 VCE 1

Consider the rhombus $OABC$ shown below, where $\vec{OA} = a\vec{i}$ and $\vec{OC} = \vec{i} + \vec{j} + \vec{k}$, and a is a positive real constant.



- i. Find a . (1 mark)
- ii. Show that the diagonals of the rhombus $OABC$ are perpendicular. (2 marks)

10. Vectors, EXT2 V1 2019 SPEC1-N 5

A triangle has vertices $A(\sqrt{3} + 1, -2, 4)$, $B(1, -2, 3)$ and $C(2, -2, \sqrt{3} + 3)$.

- i. Find angle $\angle ABC$ (3 marks)
- ii. Find the area of the triangle. (2 marks)

11. Vectors, EXT2 V1 2023 HSC 11d

The quadrilaterals $ABCD$ and $ABEF$ are parallelograms.

By considering $\vec{AB}$, show that $CDFE$ is also a parallelogram. (2 marks)

12. Vectors, EXT2 V1 EQ-Bank 10

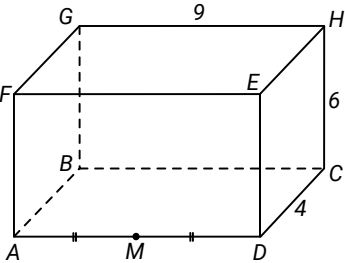
Given $\vec{a} = (3, -2, 1)$ and $\vec{b} = (4, 3, -4)$, verify numerically that

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = x_1 x_2 + y_1 y_2 + z_1 z_2$$

where $\vec{a} = (x_1, y_1, z_1)$ and $\vec{b} = (x_2, y_2, z_2)$ (4 marks)

13. Vectors, EXT2 V1 EQ-Bank 13

In a rectangular prism, M is the midpoint of AD .



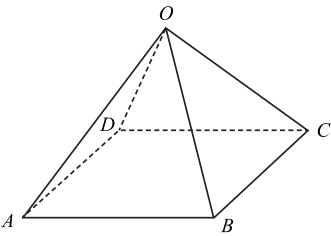
Use vector methods to find

- i. $\angle HBD$ to 1 decimal place (2 marks)
- ii. $\angle HBM$ to 1 decimal place (2 marks)

14. Vectors, EXT2 V1 EQ-Bank 8

Classify the triangle formed by joining the points $A(3, 1, 0)$, $B(-2, 4, 3)$ and $C(3, 3, -2)$. (4 marks)

15. Vectors, EXT2, V1 SM-Bank 16

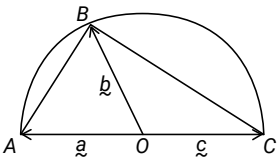


Let $OABCD$ be a right square pyramid where $\vec{a} = \vec{OA}$, $\vec{b} = \vec{OB}$, $\vec{c} = \vec{OC}$ and $\vec{d} = \vec{OD}$.

Show that $\vec{a} + \vec{c} = \vec{b} + \vec{d}$. (3 marks)

16. Vectors, EXT2 V1 EQ-Bank 15

Point B sits on the arc of a semi-circle with diameter AC .

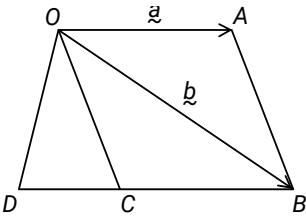


Using vectors, show $\angle ABC$ is a right angle. (2 marks)

17. Vectors, EXT2 V1 SM-Bank 20

$OABD$ is a trapezium in which $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$.

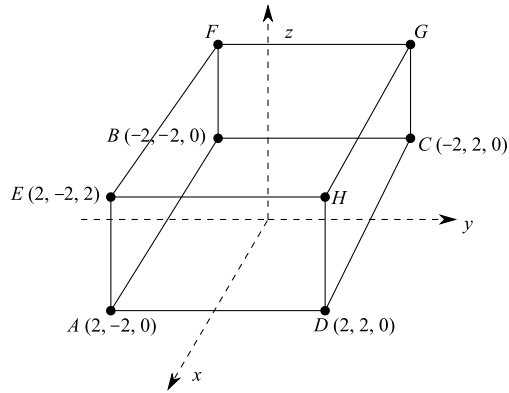
OC is parallel to AB and $DC:CB = 1:2$



Using vectors, express $\vec{DA}$ in terms of $\vec{a}$ and $\vec{b}$. (3 marks)

18. Vectors, EXT2 V1 SM-Bank 22

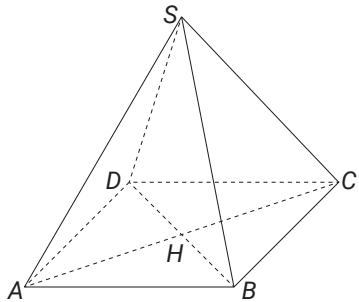
$ABCDEFGH$ are the vertices of a rectangular prism.



- Show that the internal diagonals of the prism, $\overrightarrow{AG}$ and $\overrightarrow{DF}$, intersect. (2 marks)
- Calculate the acute angle, θ , between the diagonals, to the nearest minute. (2 marks)

19. Vectors, EXT2 V1 2021 HSC 12e

The diagram shows the pyramid $ABCD S$ where $ABCD$ is a square. The diagonals of the square bisect each other at H .



- Show that $\overrightarrow{HA} + \overrightarrow{HB} + \overrightarrow{HC} + \overrightarrow{HD} = \mathbf{0}$ (1 mark)

Let G be the point such that $\overrightarrow{GA} + \overrightarrow{GB} + \overrightarrow{GC} + \overrightarrow{GD} + \overrightarrow{GS} = \mathbf{0}$.

- Using part (i), or otherwise, show that $4\overrightarrow{GH} + \overrightarrow{GS} = \mathbf{0}$. (2 marks)

- Find the value of λ such that $\overrightarrow{HG} = \lambda \overrightarrow{HS}$ (1 mark)

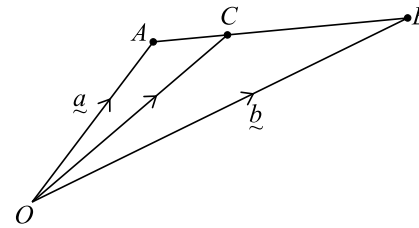
20. Vectors, EXT2 V1 2019 SPEC2 4

The base of a pyramid is the parallelogram $ABCD$ with vertices at points $A(2, -1, 3)$, $B(4, -2, 1)$, $C(a, b, c)$ and $D(4, 3, -1)$. The apex (top) of the pyramid is located at $P(4, -4, 9)$.

- Find the values of a , b and c . (2 marks)
- Find the cosine of the angle between the vectors $\overrightarrow{AB}$ and $\overrightarrow{AD}$. (2 marks)
- Find the area of the base of the pyramid. (2 marks)

21. Vectors, EXT2 V1 2020 HSC 15b

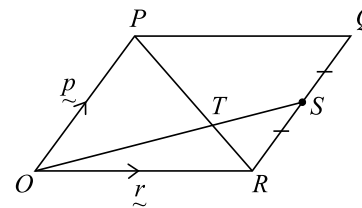
The point C divides the interval AB so that $\frac{CB}{AC} = \frac{m}{n}$. The position vectors of A and B are $\tilde{a}$ and $\tilde{b}$ respectively, as shown in the diagram.



- Show that $\overrightarrow{AC} = \frac{n}{m+n}(\tilde{b} - \tilde{a})$. (2 marks)

- Prove that $\overrightarrow{OC} = \frac{m}{m+n}\tilde{a} + \frac{n}{m+n}\tilde{b}$. (1 mark)

Let $OPQR$ be a parallelogram with $\overrightarrow{OP} = \tilde{p}$ and $\overrightarrow{OR} = \tilde{r}$. The point S is the midpoint of QR and T is the intersection of PR and OS , as shown in the diagram.

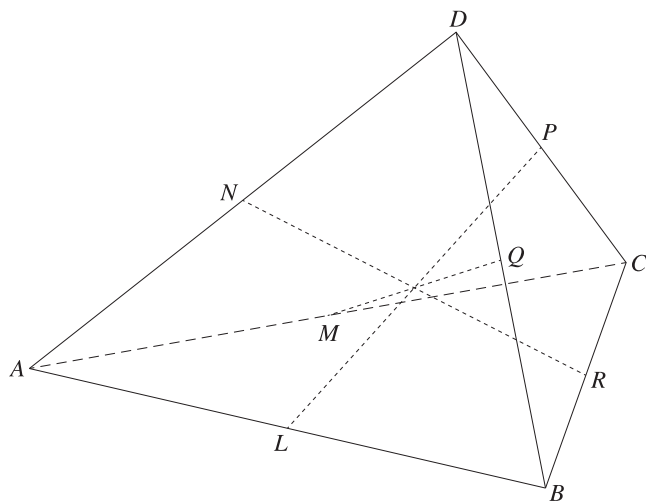


- Show that $\overrightarrow{OT} = \frac{2}{3}\tilde{r} + \frac{1}{3}\tilde{p}$. (3 marks)

- Using parts (ii) and (iii), or otherwise, prove that T is the point that divides the interval PR in the ratio 2 : 1. (1 mark)

22. Vectors, EXT2 V1 2023 HSC 15b

On the triangular pyramid $ABCD$, L is the midpoint of AB , M is the midpoint of AC , N is the midpoint of AD , P is the midpoint of CD , Q is the midpoint of BD and R is the midpoint of BC .



Let $\tilde{b} = \overrightarrow{AB}$, $\tilde{c} = \overrightarrow{AC}$ and $\tilde{d} = \overrightarrow{AD}$.

i. Show that $\overrightarrow{LP} = \frac{1}{2}(-\tilde{b} + \tilde{c} + \tilde{d})$. (1 mark)

ii. It can be shown that

$$\overrightarrow{MQ} = \frac{1}{2}(\tilde{b} - \tilde{c} + \tilde{d}) \text{ and}$$

$$\overrightarrow{NR} = \frac{1}{2}(\tilde{b} + \tilde{c} - \tilde{d}). \text{ (Do NOT prove these.)}$$

Prove that

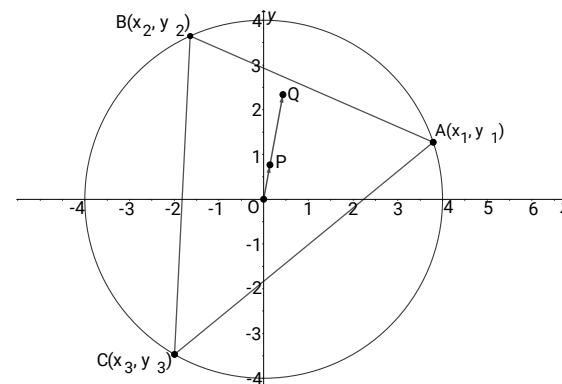
$$\begin{aligned} |\overrightarrow{AB}|^2 + |\overrightarrow{AC}|^2 + |\overrightarrow{AD}|^2 + |\overrightarrow{BC}|^2 + |\overrightarrow{BD}|^2 + |\overrightarrow{CD}|^2 \\ = 4 \left(|\overrightarrow{LP}|^2 + |\overrightarrow{MQ}|^2 + |\overrightarrow{NR}|^2 \right) \end{aligned} \quad (3 \text{ marks})$$

23. Vectors, EXT2 V1 EQ-Bank 11

Point O is the circumcentre of triangle ABC which is the centre of the circle that passes through each of its vertices.

Point P is the centroid of triangle ABC where the bisectors of each angle within the triangle intersect.

Point Q is such that $\overrightarrow{OQ} = 3\overrightarrow{OP}$.



Prove that $\overrightarrow{CQ} \perp \overrightarrow{AB}$ (5 marks)

24. Vectors, EXT2 V1 2021 HSC 16a

i. The point $P(x, y, z)$ lies on the sphere of radius 1 centred at the origin O .

Using the position vector of P , $\overrightarrow{OP} = x\tilde{i} + y\tilde{j} + z\tilde{k}$, and the triangle inequality, or otherwise, show that $|x| + |y| + |z| \geq 1$. (2 marks)

ii. Given the vectors $\tilde{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\tilde{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, show that

$$|a_1b_1 + a_2b_2 + a_3b_3| \leq \sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}. \quad (3 \text{ marks})$$

iii. As in part (i), the point $P(x, y, z)$ lies on the sphere of radius 1 centred at the origin O .

Using part (ii), or otherwise, show that $|x| + |y| + |z| \leq \sqrt{3}$. (2 marks)

25. Vectors, EXT2 V1 2022 HSC 14a

- i. The two non-parallel vectors $\vec{u}$ and $\vec{v}$ satisfy $\lambda\vec{u} + \mu\vec{v} = \vec{0}$ for some real numbers λ and μ .

Show that $\lambda = \mu = 0$. (2 marks)

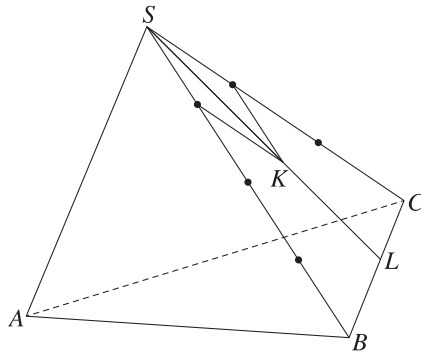
- ii. The two non-parallel vectors $\vec{u}$ and $\vec{v}$ satisfy $\lambda_1\vec{u} + \mu_1\vec{v} = \lambda_2\vec{u} + \mu_2\vec{v}$ for some real numbers $\lambda_1, \lambda_2, \mu_1$ and μ_2 .

Using part (i), or otherwise, show that $\lambda_1 = \lambda_2$ and $\mu_1 = \mu_2$. (1 mark)

The diagram below shows the tetrahedron with vertices A, B, C and S .

The point K is defined by $\vec{SK} = \frac{1}{4}\vec{SB} + \frac{1}{3}\vec{SC}$, as shown in the diagram.

The point L is the point of intersection of the straight lines SK and BC .



- iii. Using part (ii), or otherwise, determine the position of L by showing that $\vec{BL} = \frac{4}{7}\vec{BC}$. (2 marks)

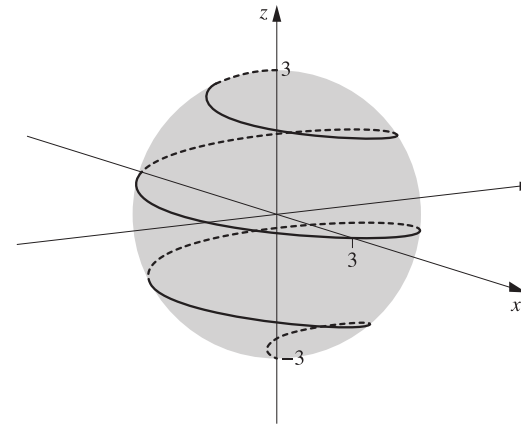
- iv. The point P is defined by $\vec{AP} = -6\vec{AB} - 8\vec{AC}$.

Does P lie on the line AL ? Justify your answer. (2 marks)

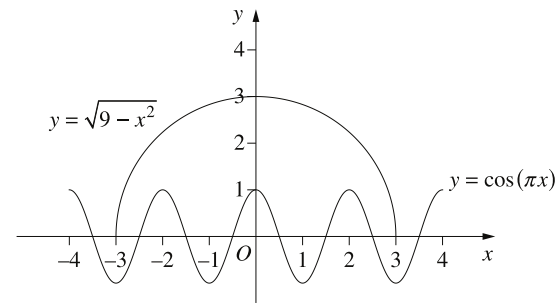
26. Vectors, EXT2 V1 2023 HSC 15c

A curve $\mathcal{C}$ spirals 3 times around the sphere centred at the origin and with radius 3, as shown.

A particle is initially at the point $(0, 0, -3)$ and moves along the curve $\mathcal{C}$ on the surface of the sphere, ending at the point $(0, 0, 3)$.



By using the diagram below, which shows the graphs of the functions $f(x) = \cos(\pi x)$ and $g(x) = \sqrt{9 - x^2}$, and considering the graph $y = f(x)g(x)$, give a possible set of parametric equations that describe the curve $\mathcal{C}$. (3 marks)



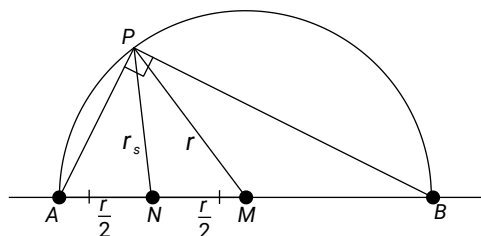
Worked Solutions

1. Vectors, EXT2 V1 2021 HSC 1 MC

$$\begin{aligned}\vec{AB} + \vec{CQ} &= \vec{CD} + \vec{DP} \\ &= \vec{CP} \\ \Rightarrow B\end{aligned}$$

2. Vectors, EXT2 V1 2022 HSC 9 MC

Diagram below is a 2-D sliced image of the 3-D geometry:



$$\vec{AP} \cdot \vec{BP} = 0.$$

$$r = \frac{|\vec{AB}|}{2}$$

$$S_2 \text{ includes } N \text{ where } |\vec{AN}| = |\vec{MN}| = \frac{r}{2}$$

Let $r_s =$ radius of S

Point P is intersection of S_1 and S_2

By Pythagoras (see diagram):

$$r_s^2 = r^2 - \left(\frac{r}{2}\right)^2$$

$$= \frac{3r^2}{4}$$

$$r_s = \frac{\sqrt{3}}{2} \times r$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{|\vec{AB}|}{2}$$

$$= \frac{\sqrt{3}|\vec{AB}|}{4}$$

$$\Rightarrow D$$

◆◆ Mean mark 20%.

Worked Solutions

3. Vectors, EXT2 V1 2023 HSC 10 MC

By elimination

Option A:

If $\hat{a}$ and $\hat{b}$ are the two unit vectors, $\hat{a} \cdot \hat{b} = |\hat{a}| |\hat{b}| \cos \theta = \cos \theta$

$$-1 \leq \cos \theta \leq 1 \Rightarrow -1 \leq \hat{a} \cdot \hat{b} \leq 1$$

$$\text{Given } \hat{a} \cdot \hat{b} = 1 \Rightarrow \hat{a} = \hat{b} \Rightarrow \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a}$$

$$\text{Contradicts } \hat{b} \cdot \hat{c} = 2 \text{ and } \hat{c} \cdot \hat{a} = 3$$

Similar reasoning rules out any pair satisfying all conditions (eliminate A).

Option C: If $\hat{a} = 0$, $\hat{a} \cdot \hat{b} = 0 \neq 1$ (eliminate C).

Option D:

Consider the vectors below that satisfy the conditions,

$$\hat{a} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \hat{b} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{c} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

However, $r\hat{a} + s\hat{b} + t\hat{c} = 0$ requires $r = s = t = 0$ which are not positive constants (eliminate D).

$$\Rightarrow B$$

◆◆ Mean mark 15%.

4. Vectors, EXT2 V1 EQ-Bank 14

Method 1

Let $\vec{c}$ = position vector of the dividing point

$$\begin{aligned}\vec{c} &= \begin{pmatrix} 7 \\ -3 \\ 0 \end{pmatrix} + \frac{2}{5} \begin{pmatrix} 2-7 \\ 2-(-3) \\ -10-0 \end{pmatrix} \\ &= \begin{pmatrix} 7 \\ -3 \\ 0 \end{pmatrix} + \begin{pmatrix} -2 \\ 2 \\ -4 \end{pmatrix} \\ &= \begin{pmatrix} 5 \\ -1 \\ -4 \end{pmatrix}\end{aligned}$$

Method 2

$$\begin{aligned}\vec{c} &= \begin{pmatrix} 2 \\ 2 \\ -10 \end{pmatrix} + \frac{3}{5} \begin{pmatrix} 7-2 \\ -3-2 \\ 0-(-10) \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 2 \\ -10 \end{pmatrix} + \begin{pmatrix} 3 \\ -3 \\ 6 \end{pmatrix} \\ &= \begin{pmatrix} 5 \\ -1 \\ -4 \end{pmatrix}\end{aligned}$$

5. Vectors, EXT2 V1 EQ-Bank 5

Method 1

Let $\vec{c}$ = position vector of the midpoint

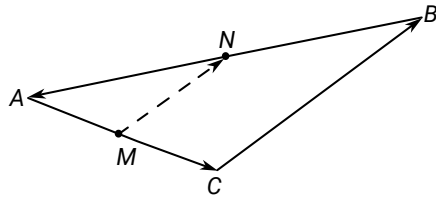
$$\begin{aligned}\vec{c} &= \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 5-3 \\ 4-(-2) \\ -3-1 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix}\end{aligned}$$

Method 2

$$\begin{aligned}\vec{c} &= \begin{pmatrix} 5 \\ 4 \\ -3 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 3-5 \\ -2-4 \\ 1-(-3) \end{pmatrix} \\ &= \begin{pmatrix} 5 \\ 4 \\ -3 \end{pmatrix} + \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix}\end{aligned}$$

6. Vectors, EXT2 V1 EQ-Bank 7

i.



$$\vec{AC} + \vec{CB} + \vec{BA} = \vec{0} \quad (\text{resultant vector ends at starting point})$$

$$\vec{CB} = -(\vec{AC} + \vec{BA})$$

Since M and N are midpoints:

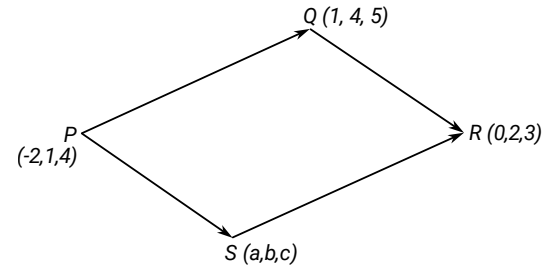
$$\begin{aligned} \vec{MN} &= -\frac{1}{2}\vec{AC} - \frac{1}{2}\vec{BA} \\ &= -\frac{1}{2}(\vec{AC} + \vec{BA}) \\ &= \frac{1}{2}\vec{CB} \quad \dots \text{ as required} \end{aligned}$$

ii. If $\vec{u} = k\vec{v}$ (k scalar) $\Rightarrow \vec{u} \parallel \vec{v}$

$$\vec{MN} = \frac{1}{2}\vec{CB} \quad (\text{see (i)})$$

$$\therefore MN \parallel CB$$

7. Vectors, EXT2 V1 EQ-Bank 9



Opposite sides of a parallelogram are equal and parallel.

$$\Rightarrow \vec{PQ} = \vec{SR}$$

$$\vec{PQ} = \begin{pmatrix} 1+2 \\ 4-1 \\ 5-4 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix}$$

$$\vec{SR} = \begin{pmatrix} -a \\ 2-b \\ 3-c \end{pmatrix}$$

Equating coordinates:

$$-a = 3 \Rightarrow a = -3$$

$$2-b = 1 \Rightarrow b = 1$$

$$3-c = 1 \Rightarrow c = 2$$

8. Vectors, EXT2 V1 SM-Bank 23

i. $A(3, 0, 0), \quad G(0, 3, 3)$

$$\overrightarrow{AG} = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix}$$

$$\left| \overrightarrow{AG} \right| = \sqrt{9 + 9 + 9} \\ = 3\sqrt{3} \text{ units}$$

ii. $H(3, 3, 3)$

$$\overrightarrow{BH} = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$$

$$\overrightarrow{AG} \cdot \overrightarrow{BH} = \left| \overrightarrow{AG} \right| \cdot \left| \overrightarrow{BH} \right| \cos\theta$$

$$\begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} = \sqrt{9 + 9 + 9} \cdot \sqrt{9 + 9 + 9} \cos\theta$$

$$-9 + 9 + 9 = 27\cos\theta$$

$$\cos\theta = \frac{1}{3}$$

$$\theta = 70.52\dots$$

$$= 70^\circ 32'$$

9. Vectors, EXT2 V1 2015 VCE 1

i. $\left| \overrightarrow{OA} \right| = \left| \overrightarrow{OC} \right|,$

$$\therefore a = \sqrt{1^2 + 1^2 + 1^2} \\ = \sqrt{3}$$

ii. $\overrightarrow{CB} = \overrightarrow{OA}, \quad \overrightarrow{AB} = \overrightarrow{OC}$

$$\overrightarrow{OB} = \overrightarrow{OC} + \overrightarrow{CB}$$

$$= \underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k} + \sqrt{3}\underset{\sim}{i}$$

$$= \left(\sqrt{3} + 1 \right) \underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$$

$$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC}$$

$$= -\sqrt{3}\underset{\sim}{i} + \underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$$

$$= \left(1 - \sqrt{3} \right) \underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$$

$$\overrightarrow{AC} \cdot \overrightarrow{OB} = \left(1 + \sqrt{3} \right) \left(1 - \sqrt{3} \right) + 1 + 1$$

$$= 1 - 3 + 1 + 1$$

$$= 0$$

$$\therefore \overrightarrow{AC} \perp \overrightarrow{OB}$$

MARKER'S COMMENT: Vector notation was poor in many answers.

10. Vectors, EXT2 V1 2019 SPEC1-N 5

$$\text{i. } \overrightarrow{BA} = \begin{pmatrix} \sqrt{3}+1 \\ -2 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} \sqrt{3} \\ 0 \\ 1 \end{pmatrix}$$

$$\overrightarrow{BC} = \begin{pmatrix} 2 \\ -2 \\ \sqrt{3}+3 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ \sqrt{3} \end{pmatrix}$$

$$\cos \angle ABC = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{\left| \overrightarrow{BA} \right| \left| \overrightarrow{BC} \right|}$$

$$= \frac{2\sqrt{3}}{\sqrt{4}\sqrt{4}}$$

$$= \frac{\sqrt{3}}{2}$$

$$\therefore \angle ABC = \frac{\pi}{6}$$

$$\text{ii. Area} = \frac{1}{2} \cdot \left| \overrightarrow{BA} \right| \left| \overrightarrow{BC} \right| \sin \angle ABC$$

$$= \frac{1}{2} \times 2 \times 2 \times \sin \frac{\pi}{6}$$

$$= 1 \text{ u}^2$$

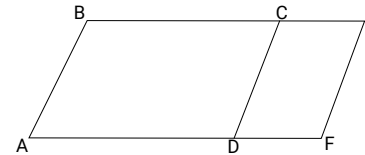
11. Vectors, EXT2 V1 2023 HSC 11d

Parallelogram $\Rightarrow$ Show opposite sides are equal

$$\left| \overrightarrow{AB} \right| = \left| \overrightarrow{CD} \right| \quad (ABCD \text{ is a parallelogram})$$

$$\left| \overrightarrow{AB} \right| = \left| \overrightarrow{EF} \right| \quad (ABEF \text{ is a parallelogram})$$

$$\Rightarrow \left| \overrightarrow{CD} \right| = \left| \overrightarrow{EF} \right|$$



$$\left| \overrightarrow{CE} \right| = \left| \overrightarrow{BE} \right| - \left| \overrightarrow{BC} \right|$$

Similarly,

$$\left| \overrightarrow{DF} \right| = \left| \overrightarrow{AF} \right| - \left| \overrightarrow{AD} \right|$$

$$\text{Since } \left| \overrightarrow{BE} \right| = \left| \overrightarrow{AF} \right| \text{ and } \left| \overrightarrow{BC} \right| = \left| \overrightarrow{AD} \right|$$

($ABCD$ and $ABEF$ are parallelograms)

$$\Rightarrow \left| \overrightarrow{DF} \right| = \left| \overrightarrow{CE} \right|$$

$\therefore CDFE$ is a parallelogram

12. Vectors, EXT2 V1 EQ-Bank 10

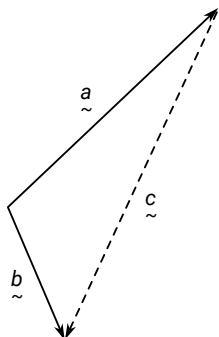
Scalar Product result 1:

$$\begin{aligned}\vec{a} \cdot \vec{b} &= x_1x_2 + y_1y_2 + z_1z_2 \\ &= 3 \times 4 + (-2) \times 3 + 1 \times (-4) \\ &= 2\end{aligned}$$

Scalar Product result 2:

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\text{Let } \vec{c} = \vec{b} - \vec{a}$$



$$\vec{c} = \begin{pmatrix} 4 \\ 3 \\ -4 \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ -5 \end{pmatrix}$$

$$|\vec{c}| = \sqrt{1^2 + 5^2 + (-5)^2} = \sqrt{51}$$

$$|\vec{a}| = \sqrt{3^2 + (-2)^2 + 1^2} = \sqrt{14}$$

$$|\vec{b}| = \sqrt{4^2 + 3^2 + (-4)^2} = \sqrt{41}$$

Using cosine rule: $c^2 = a^2 + b^2 - 2ab \cos C$

$$\Rightarrow ab \cos C = \frac{a^2 + b^2 - c^2}{2}$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$= \frac{|\vec{a}|^2 + |\vec{b}|^2 - |\vec{c}|^2}{2}$$

$$= \frac{14 + 41 - 51}{2}$$

$$= 2$$

$$\therefore \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = x_1x_2 + y_1y_2 + z_1z_2$$

13. Vectors, EXT2 V1 EQ-Bank 13

i. Consider the position vectors of points using B as origin:

$$\vec{BH} = \begin{pmatrix} 0 \\ 9 \\ 6 \end{pmatrix}, \quad |\vec{BH}| = \sqrt{9^2 + 6^2} = \sqrt{117}$$

$$\vec{BD} = \begin{pmatrix} 4 \\ 9 \\ 0 \end{pmatrix}, \quad |\vec{BD}| = \sqrt{4^2 + 9^2} = \sqrt{97}$$

$$\begin{aligned}\cos \angle HBD &= \frac{\vec{BH} \cdot \vec{BD}}{|\vec{BH}| \cdot |\vec{BD}|} \\ &= \frac{0 \times 4 + 9 \times 9 + 6 \times 0}{\sqrt{117} \times \sqrt{97}}\end{aligned}$$

$$\begin{aligned}\angle HBD &= \cos^{-1} \left(\frac{81}{\sqrt{117} \times \sqrt{97}} \right) \\ &= 40.5^\circ\end{aligned}$$

$$\text{ii. } \vec{BM} = \begin{pmatrix} 4 \\ \frac{9}{2} \\ 0 \end{pmatrix}, \quad |\vec{BM}| = \sqrt{4^2 + \left(\frac{9}{2}\right)^2} = \sqrt{\frac{145}{4}}$$

$$\begin{aligned}\cos \angle HBM &= \frac{\vec{BH} \cdot \vec{BM}}{|\vec{BH}| \cdot |\vec{BM}|} \\ &= \frac{0 \times 4 + 9 \times \frac{9}{2} + 6 \times 0}{\sqrt{99} \times \sqrt{\frac{145}{4}}}\end{aligned}$$

$$\begin{aligned}\angle HBM &= \cos^{-1} \left(\frac{40.5}{\sqrt{99} \times \sqrt{\frac{145}{4}}} \right) \\ &= 47.5^\circ\end{aligned}$$

14. Vectors, EXT2 V1 EQ-Bank 8

Calculating the side lengths:

$$|AB| = \sqrt{(3+2)^2 + (1-4)^2 + (0-3)^2} = \sqrt{43}$$

$$|BC| = \sqrt{(-2-3)^2 + (4-3)^2 + (3+2)^2} = \sqrt{51}$$

$$|AC| = \sqrt{(3-3)^2 + (1-3)^2 + (0+2)^2} = \sqrt{8}$$

$\Rightarrow$ Triangle is scalene.

From above,

$$|BC|^2 = |AB|^2 + |AC|^2 \quad (\angle BAC = 90^\circ)$$

Consider scalar product of direction vectors:

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2-3 \\ 4-1 \\ 3-0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 3 \\ 3 \end{pmatrix}$$

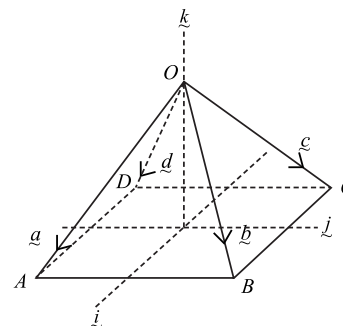
$$\overrightarrow{AC} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 3-3 \\ 3-1 \\ -2-0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix}$$

$$\overrightarrow{AB} \cdot \overrightarrow{AC} = \begin{pmatrix} -5 \\ 3 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix} = 0 + 6 - 6 = 0$$

$$\therefore \overrightarrow{AB} \perp \overrightarrow{AC}$$

$\therefore \triangle ABC$ is a right-angled (scalene) triangle with a right-angle at A.

15. Vectors, EXT2, V1 SM-Bank 16



Let $A = (p, -p, -k)$,

$$\vec{a} = \overrightarrow{OA} = p\vec{i} - p\vec{j} - q\vec{k}$$

$$\underline{b} = \overrightarrow{OB} = p_i \underline{i} + p_j \underline{j} - q_k \underline{k}$$

$$\vec{c} = \overrightarrow{OC} = -p_i + p_j - q_k$$

$$\vec{d} = \overrightarrow{OA} = -\vec{p}_i - \vec{p}_j - \vec{q}_k$$

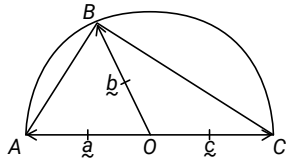
$$\underset{\sim}{a} + \underset{\sim}{c} = -2\underset{\sim}{qk}$$

$$\underset{\sim}{b} + \underset{\sim}{d} = - \underset{\sim}{2qk}$$

$$\therefore \underset{\sim}{a} + \underset{\sim}{c} = \underset{\sim}{b} + \underset{\sim}{d}$$

16. Vectors, EXT2 V1 EQ-Bank 15

Let $\vec{OA} = \underline{a}$, and $\vec{OC} = \underline{c}$



Prove $\vec{AB} \perp \vec{BC}$

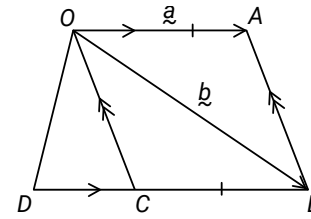
$$|\underline{a}| = |\underline{b}| = |\underline{c}| \quad (\text{radii})$$

$$\underline{c} = -\underline{a}$$

$$\begin{aligned} \vec{AB} \cdot \vec{BC} &= (\underline{b} - \underline{a})(\underline{c} - \underline{b}) \\ &= \underline{b} \cdot \underline{c} - |\underline{b}|^2 - \underline{a} \cdot \underline{c} + \underline{a} \cdot \underline{b} \\ &= \underline{b} \cdot \underline{c} - |\underline{b}|^2 + \underline{c} \cdot \underline{c} - \underline{c} \cdot \underline{b} \\ &= |\underline{c}|^2 - |\underline{b}|^2 \\ &= 0 \end{aligned}$$

$\therefore \angle ABC$ is a right angle.

17. Vectors, EXT2 V1 SM-Bank 20



$$\vec{DA} = \vec{DB} + \vec{BA}$$

$$\vec{BA} = \underline{a} - \underline{b}$$

Since $OC \parallel AB$ and $OA \parallel CB$
 $\Rightarrow OACB$ is a parallelogram

$$\vec{OA} = \vec{CB} = \underline{a}$$

$$\vec{DC} = \frac{1}{2} \underline{a}$$

$$\begin{aligned} \vec{DB} &= \vec{DC} + \vec{CB} \\ &= \frac{3}{2} \underline{a} \end{aligned}$$

$$\begin{aligned} \therefore \vec{DA} &= \frac{3}{2} \underline{a} + (\underline{a} - \underline{b}) \\ &= \frac{5}{2} \underline{a} - \underline{b} \end{aligned}$$

18. Vectors, EXT2 V1 SM-Bank 22

i. $A(2, -2, 0), G(-2, 2, 2)$

$D(2, 2, 0), F(-2, -2, 2)$

$$\text{Midpoint } AG = \left(\frac{\frac{1}{2}(2-2)}{\frac{1}{2}(-2+2)} \right) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{Midpoint } DF = \left(\frac{\frac{1}{2}(2-2)}{\frac{1}{2}(2-2)} \right) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Since midpoints are the same, AG and DF intersect.

ii. $\vec{AG} = \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 4 \\ 2 \end{pmatrix}$

$$\vec{DF} = \begin{pmatrix} -2 \\ -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \\ 2 \end{pmatrix}$$

$$\vec{AG} \cdot \vec{DF} = \left| \vec{AG} \right| \cdot \left| \vec{DF} \right| \cos \theta$$

$$\begin{pmatrix} -4 \\ 4 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ -4 \\ 2 \end{pmatrix} = \sqrt{36} \sqrt{36} \cos \theta$$

$$16 - 16 + 4 = 36 \cos \theta$$

$$\cos \theta = \frac{1}{9}$$

$$\theta = 83.62 \dots$$

$$= 83^\circ 37'$$

19. Vectors, EXT2 V1 2021 HSC 12e

i. Since diagonal $\vec{AC}$ is bisected by H:

$$\vec{HA} = -\vec{HC}$$

Similarly for diagonal $\vec{BD}$

$$\vec{HB} = -\vec{HD}$$

$$\therefore \vec{HA} + \vec{HB} + \vec{HC} + \vec{HD} = -\vec{HC} - \vec{HD} + \vec{HC} + \vec{HD} = 0$$

ii. $\vec{GA} = \vec{GH} + \vec{HA}, \vec{GB} = \vec{GH} + \vec{HB}$
 $\vec{GC} = \vec{GH} + \vec{HC}, \vec{GD} = \vec{GH} + \vec{HD}$

$$\begin{aligned} \vec{GA} + \vec{GB} + \vec{GC} + \vec{GD} + \vec{GS} &= \vec{GH} + \vec{HA} + \vec{GH} + \vec{HB} + \vec{GH} + \vec{HC} + \vec{GH} + \vec{HD} + \vec{GS} \\ &= 4\vec{GH} + (\vec{HA} + \vec{HB} + \vec{HC} + \vec{HD} + \vec{GS}) \\ &= 4\vec{GH} + \vec{GS} \end{aligned}$$

$$\vec{GA} + \vec{GB} + \vec{GC} + \vec{GD} + \vec{GS} = 0 \quad (\text{given})$$

$$\therefore 4\vec{GH} + \vec{GS} = 0$$

iii. $\vec{HS} = \vec{HG} + \vec{GS}$
 $\vec{GS} = \vec{HS} + \vec{GH}$

Using part (ii):

$$4\vec{GH} + \vec{HS} + \vec{GH} = 0$$

$$5\vec{GH} = \vec{SH}$$

$$\vec{GH} = \frac{1}{5} \vec{SH}$$

$$\vec{HG} = \frac{1}{5} \vec{HS}$$

$$\therefore \lambda = \frac{1}{5}$$

♦ Mean mark 48%.

20. Vectors, EXT2 V1 2019 SPEC2 4

$$\begin{aligned} \text{i. } \vec{AB} &= (4-2)\hat{i} + (-2+1)\hat{j} + (1-3)\hat{k} \\ &= 2\hat{i} - \hat{j} - 2\hat{k} \end{aligned}$$

Since $ABCD$ is a parallelogram $\Rightarrow \vec{AB} = \vec{DC}$

$$\vec{DC} = (a-4)\hat{i} + (b-3)\hat{j} + (c+1)\hat{k}$$

$$a-4 = 2 \Rightarrow a = 6$$

$$b-3 = -1 \Rightarrow b = 2$$

$$c+1 = -2 \Rightarrow c = -3$$

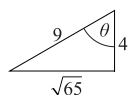
$$\text{ii. } \vec{AB} = 2\hat{i} - \hat{j} - 2\hat{k}$$

$$\vec{AD} = 2\hat{i} + 4\hat{j} - 4\hat{k}$$

$$\begin{aligned} \cos \angle BAD &= \frac{\vec{AB} \cdot \vec{AD}}{|\vec{AB}| \cdot |\vec{AD}|} \\ &= \frac{4-4+8}{\sqrt{4+1+4} \cdot \sqrt{4+16+16}} \\ &= \frac{4}{9} \end{aligned}$$

$$\text{iii. } \frac{1}{2} \times \text{Area}_{ABCD} = \frac{1}{2} absinc$$

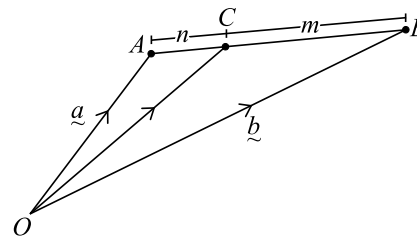
$$\text{Area}_{ABCD} = |\vec{AB}| \cdot |\vec{AD}| \cdot \sin\left(\cos^{-1} \frac{4}{9}\right)$$



$$\begin{aligned} \therefore \text{Area}_{ABCD} &= 3 \cdot 6 \cdot \frac{\sqrt{65}}{9} \\ &= 2\sqrt{65} \text{ u}^2 \end{aligned}$$

21. Vectors, EXT2 V1 2020 HSC 15b

i.

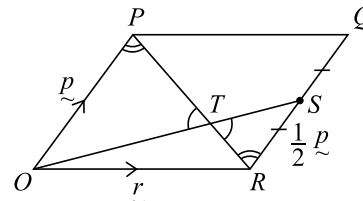


$$\frac{\vec{AC}}{\vec{AB}} = \frac{n}{m+n}$$

$$\begin{aligned} \vec{AC} &= \frac{n}{m+n} \cdot \vec{AB} \\ &= \frac{n}{m+n} (\vec{b} - \vec{a}) \end{aligned}$$

$$\begin{aligned} \text{ii. } \vec{OC} &= \vec{OA} + \vec{AC} \\ &= \vec{a} + \frac{n}{m+n} (\vec{b} - \vec{a}) \\ &= \vec{a} - \frac{n}{m+n} \vec{a} + \frac{n}{m+n} \vec{b} \\ &= \left(1 - \frac{n}{m+n}\right) \vec{a} + \frac{n}{m+n} \vec{b} \\ &= \left(\frac{m+n-n}{m+n}\right) \vec{a} + \frac{n}{m+n} \vec{b} \\ &= \frac{m}{m+n} \vec{a} + \frac{n}{m+n} \vec{b} \end{aligned}$$

$$\text{iii. Show } \vec{OT} = \frac{2}{3} \vec{r} + \frac{1}{3} \vec{p}$$



Consider $\triangle PTO$ and $\triangle RTS$:

$\angle PTO = \angle RTS$ (vertically opposite)

$\angle OPT = \angle SRT$ (vertically opposite)

♦ Mean mark part (iii) 50%.

$\therefore \Delta PTO \parallel \Delta RTS$ (equiangular)

$$OT:TS = OP:SR = 2:1$$

(corresponding sides in the same ratio)

$$\frac{\overrightarrow{OT}}{\overrightarrow{OS}} = \frac{2}{3}$$

$$\begin{aligned}\overrightarrow{OT} &= \frac{2}{3}\overrightarrow{OS} \\ &= \frac{2}{3}\left(\tilde{r} + \frac{1}{2}\tilde{p}\right) \\ &= \frac{2}{3}\tilde{r} + \frac{1}{3}\tilde{p}\end{aligned}$$

iv. Let $\overrightarrow{OT}$ divide PR so that $\frac{TR}{PT} = \frac{m}{n}$

Using part (ii):

$$\overrightarrow{OT} = \frac{m}{m+n}\tilde{p} + \frac{n}{m+n}\tilde{c}$$

$$\overrightarrow{OT} = \frac{1}{3}\tilde{p} + \frac{2}{3}\tilde{r} \quad (\text{part (iii)})$$

$$\frac{m}{m+n} = \frac{1}{3}, \frac{n}{m+n} = \frac{2}{3}$$

$$\Rightarrow m = 1, n = 2$$

$\therefore T$ divides PR in ratio $2:1$.

Mean mark part (iv) 51%.

22. Vectors, EXT2 V1 2023 HSC 15b

$$\begin{aligned}\text{i. } \overrightarrow{LP} &= \overrightarrow{LA} + \overrightarrow{AC} + \overrightarrow{CP} \\ &= \frac{1}{2}\overrightarrow{BA} + \overrightarrow{AC} + \frac{1}{2}\overrightarrow{CD} \\ &= -\frac{1}{2}\tilde{b} + \tilde{c} + \frac{1}{2}(\tilde{d} - \tilde{c}) \\ &= -\frac{1}{2}\tilde{b} + \tilde{c} + \frac{1}{2}\tilde{d} - \frac{1}{2}\tilde{c} \\ &= \frac{1}{2}(-\tilde{b} + \tilde{c} + \tilde{d})\end{aligned}$$

$$\text{ii. } \overrightarrow{MQ} = \frac{1}{2}(\tilde{b} - \tilde{c} + \tilde{d})$$

$$\overrightarrow{NR} = \frac{1}{2}(\tilde{b} + \tilde{c} - \tilde{d})$$

♦♦ Mean mark (ii) 34%.

$$\begin{aligned}\text{RHS} &= 4\left(\left|\overrightarrow{LP}\right|^2 + \left|\overrightarrow{MQ}\right|^2 + \left|\overrightarrow{NR}\right|^2\right) \\ &= 4\left(\overrightarrow{LP} \cdot \overrightarrow{LP} + \overrightarrow{MQ} \cdot \overrightarrow{MQ} + \overrightarrow{NR} \cdot \overrightarrow{NR}\right) \\ &= 4\left(\frac{1}{4}(-\tilde{b} + \tilde{c} + \tilde{d}) \cdot (-\tilde{b} + \tilde{c} + \tilde{d}) + \right. \\ &\quad \left. \frac{1}{4}(\tilde{b} - \tilde{c} + \tilde{d}) \cdot (\tilde{b} - \tilde{c} + \tilde{d}) + \right. \\ &\quad \left. \frac{1}{4}(\tilde{b} + \tilde{c} - \tilde{d}) \cdot (\tilde{b} + \tilde{c} - \tilde{d})\right) \\ &= (\tilde{c} + (\tilde{d} - \tilde{b})) \cdot (\tilde{c} + (\tilde{d} - \tilde{b})) + (\tilde{d} + (\tilde{b} - \tilde{c})) \cdot (\tilde{d} + (\tilde{b} - \tilde{c})) + \\ &\quad (\tilde{b} + (\tilde{c} - \tilde{d})) \cdot (\tilde{b} + (\tilde{c} - \tilde{d})) \\ &= |\tilde{c}|^2 + 2\tilde{c}(\tilde{d} - \tilde{b}) + |\tilde{d} - \tilde{b}|^2 + \\ &\quad |\tilde{d}|^2 + 2\tilde{d}(\tilde{b} - \tilde{c}) + |\tilde{b} - \tilde{c}|^2 + \\ &\quad |\tilde{b}|^2 + 2\tilde{b}(\tilde{c} - \tilde{d}) + |\tilde{c} - \tilde{d}|^2 \\ &= |\tilde{b}|^2 + |\tilde{c}|^2 + |\tilde{d}|^2 + |\tilde{b} - \tilde{c}|^2 + |\tilde{d} - \tilde{b}|^2 + |\tilde{c} - \tilde{d}|^2 + \\ &\quad 2(\tilde{c} \cdot \tilde{d} - \tilde{c} \cdot \tilde{b} + \tilde{d} \cdot \tilde{b} - \tilde{d} \cdot \tilde{c} + \tilde{b} \cdot \tilde{c} - \tilde{b} \cdot \tilde{d}) \\ &= \left|\overrightarrow{AB}\right|^2 + \left|\overrightarrow{AC}\right|^2 + \left|\overrightarrow{AD}\right|^2 + \left|\overrightarrow{BC}\right|^2 + \left|\overrightarrow{BD}\right|^2 + \left|\overrightarrow{CD}\right|^2 + 0 \\ &= \text{LHS}\end{aligned}$$

23. Vectors, EXT2 V1 EQ-Bank 11

$$\overrightarrow{OP} = \frac{\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}}{3}$$

$$\begin{aligned}\overrightarrow{OQ} &= 3 \overrightarrow{OP} \\ &= 3 \left(\frac{\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}}{3} \right) \\ &= \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} \\ &= \begin{pmatrix} x_1 + x_2 + x_3 \\ y_1 + y_2 + y_3 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\overrightarrow{CQ} &= \overrightarrow{OQ} - \overrightarrow{OC} \\ &= \begin{pmatrix} x_1 + x_2 + x_3 \\ y_1 + y_2 + y_3 \end{pmatrix} - \begin{pmatrix} x_3 \\ y_3 \end{pmatrix} \\ &= \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} - \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \\ &= \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}\end{aligned}$$

Test $\overrightarrow{CQ} \cdot \overrightarrow{AB} = 0$:

$$\begin{aligned}\text{LHS} &= \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix} \cdot \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix} \\ &= \begin{pmatrix} (x_1 + x_2)(x_2 - x_1) \\ (y_1 + y_2)(y_2 - y_1) \end{pmatrix} \\ &= (x_2)^2 - (x_1)^2 + (y_2)^2 - (y_1)^2 \\ &= (x_2)^2 + (y_2)^2 - ((x_1)^2 + (y_1)^2) \\ &= \left| \overrightarrow{OB} \right|^2 - \left| \overrightarrow{OA} \right|^2 \\ &= OB^2 - OA^2 \quad (\text{radii}) \\ &= 0\end{aligned}$$

$$\therefore \overrightarrow{CQ} \perp \overrightarrow{AB} \dots \text{as required}$$

24. Vectors, EXT2 V1 2021 HSC 16a

i. **Triangle inequality:** $|x| + |y| \geq |x + y|$

$$\begin{aligned}|x| + |y| + |z| &= \left| \begin{pmatrix} x_i \\ y_j \\ z_k \end{pmatrix} \right| \\ &\geq \left| \begin{pmatrix} x_i + y_j \\ z_k \end{pmatrix} \right| \\ &\geq \left| \begin{pmatrix} x_i + y_j + z_k \end{pmatrix} \right| \\ &\geq 1 \quad \left(\left| \overrightarrow{OP} \right| = \left| \begin{pmatrix} x_i + y_j + z_k \end{pmatrix} \right| = 1 \right)\end{aligned}$$

♦ Mean mark (i) 50%.

ii. Using the dot product:

$$\underline{a} \cdot \underline{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta$$

♦ Mean mark (ii) 42%.

$$a_1 b_1 + a_2 b_2 + a_3 b_3 = \sqrt{a_1^2 + a_2^2 + a_3^2} \cdot \sqrt{b_1^2 + b_2^2 + b_3^2} \cdot \cos \theta$$

$$|a_1 b_1 + a_2 b_2 + a_3 b_3| = \sqrt{a_1^2 + a_2^2 + a_3^2} \cdot \sqrt{b_1^2 + b_2^2 + b_3^2} \cdot |\cos \theta|$$

$$\text{Since } -1 \leq \cos \theta \leq 1 \Rightarrow |\cos \theta| \leq 1$$

$$\therefore |a_1 b_1 + a_2 b_2 + a_3 b_3| \leq \sqrt{a_1^2 + a_2^2 + a_3^2} \cdot \sqrt{b_1^2 + b_2^2 + b_3^2}$$

iii. Using part (ii) with vectors:

$$\underline{a} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \underline{b} = \begin{pmatrix} |x| \\ |y| \\ |z| \end{pmatrix}$$

♦♦♦ Mean mark (iii) 14%.

$$| |x| + |y| + |z| | \leq \sqrt{1^2 + 1^2 + 1^2} \cdot \sqrt{x^2 + y^2 + z^2}$$

$$|x| + |y| + |z| \leq \sqrt{3}$$

25. Vectors, EXT2 V1 2022 HSC 14a

i. $\lambda \vec{u} + \mu \vec{v} = \vec{0}$

$$\lambda \vec{u} = -\mu \vec{v}$$

$$\lambda = 0 \text{ or } \vec{u} = -\left(\frac{\mu}{\lambda}\right)\vec{v} = k\vec{v} \quad (k \in \mathbb{R})$$

Since $\vec{u}$ and $\vec{v}$ are not parallel

$$\Rightarrow \lambda = \mu = 0$$

ii. $\lambda_1 \vec{u} + \mu_1 \vec{v} = \lambda_2 \vec{u} + \mu_2 \vec{v}$

$$\lambda_1 \vec{u} - \lambda_2 \vec{u} + \mu_1 \vec{v} - \mu_2 \vec{v} = \vec{0}$$

$$(\lambda_1 - \lambda_2)\vec{u} + (\mu_1 - \mu_2)\vec{v} = \vec{0}$$

Using part (i):

$$(\lambda_1 - \lambda_2) = 0 \text{ and } (\mu_1 - \mu_2) = 0$$

$$\therefore \lambda_1 = \lambda_2 \text{ and } \mu_1 = \mu_2 \dots \text{ as required}$$

iii. $\vec{BL} = \vec{\lambda BC}$

$$= \lambda(\vec{BS} + \vec{SC}) \dots (1)$$

$$\vec{BL} = \vec{BS} + \mu \vec{SK}$$

$$= -\vec{SB} + \mu\left(\frac{1}{4}\vec{SB} + \frac{1}{3}\vec{SC}\right)$$

$$= -\vec{SB} + \frac{\mu}{4}\vec{SB} + \frac{\mu}{3}\vec{SC}$$

$$= \frac{\mu}{3}\vec{SC} + \left(\frac{\mu}{4} - 1\right)\vec{SB} \dots (2)$$

Using (1) = (2):

$$\lambda \vec{BS} + \lambda \vec{SC} = \frac{\mu}{3}\vec{SC} + \left(\frac{\mu}{4} - 1\right)\vec{SB}$$

$$\frac{\mu}{3} = \lambda, \quad 1 - \frac{\mu}{4} = \lambda$$

$$\frac{\mu}{3} = 1 - \frac{\mu}{4}$$

$$\frac{4\mu + 3\mu}{12} = 1$$

$$\frac{7\mu}{12} = 1$$

$$\mu = \frac{12}{7}$$

Mean mark (ii) 56%.

♦♦♦ Mean mark (iii) 25%.

$$\lambda = \frac{\frac{12}{7}}{3} = \frac{4}{7}$$

$$\therefore \vec{BL} = \frac{4}{7}\vec{BC} \dots \text{ as required}$$

iv. $\vec{AP} = -6\vec{AB} - 8\vec{AC}$

If P lies on AL , $\exists k$ such that $\vec{AP} = k\vec{AL}$

$$-6\vec{AB} - 8\vec{AC} = k\vec{AL}$$

$$-6\vec{AB} - 8\left(\vec{AB} + \vec{BC}\right) = k\left(\vec{AB} + \vec{BL}\right)$$

$$-14\vec{AB} - 8\vec{BC} = k\left(\vec{AB} + \frac{4}{7}\vec{BC}\right) \quad (\text{see part (iii)})$$

$$-14\vec{AB} - 8\vec{BC} = k\vec{AB} + \frac{4k}{7}\vec{BC}$$

$$k = -14, \quad \frac{4k}{7} = -8 \Rightarrow k = -14$$

$\therefore P$ lies on AL .

♦♦♦ Mean mark (iv) 23%.

26. Vectors, EXT2 V1 2023 HSC 15c

Since the curve lies on a sphere with radius 3:

$$x^2 + y^2 + z^2 = 3^2$$

Considering the graph $y = \cos(\pi t)\sqrt{9 - t^2}$ (as per hint)

$$\left(\cos(\pi t)\sqrt{9 - t^2}\right)^2 + \left(\sin(\pi t)\sqrt{9 - t^2}\right)^2 + t^2 = 3^2 \dots (1)$$

Since z increases and x and y change signs

$$\Rightarrow z = t$$

In order to satisfy the equation in (1):

$$x, y \text{ must be one of } \pm \cos(\pi t)\sqrt{9 - t^2} \text{ or } \pm \sin(\pi t)\sqrt{9 - t^2}$$

At $z = 0, t = 0, x = 3$ (from graph):

$$\Rightarrow x = \cos(\pi t)\sqrt{9 - t^2}$$

At $z = 0 + \epsilon, t = 0 + \epsilon, y < 0$ (from graph):

$$\Rightarrow y = -\sin(\pi t)\sqrt{9 - t^2}$$

♦♦♦ Mean mark 22%.

